



HIMESH PRATAP SINGH

Final Year Undergraduate
Department of Electrical Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023-2027	8.3/10
Intermediate/+2	CBSE (XII)	New Kingston Senior Secondary School	2023	95.0%
Matriculation	ICSE (X)	St. Mary's Orthodox School	2021	99.0%

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 416**, at the **IICPC Quantfest Prelims 2025**, advancing to the regionals round.
- Secured **All India Rank 509**, in **KVPY (SA Stream) 2022** among 50,000 candidates, conducted by IISc Bangalore.
- Achieved **5th position among 23 IITs**, at **Inter IIT Tech Meet 13.0** for the FedEx optimisation problem statement.
- Qualified for **Indian National Astronomy Olympiad (INAO) twice**, securing **Top 43 State Rank** in 2021 and 2022.
- Awarded **Merit Certificate**, for performance in the **Indian Olympiad Qualifier for Mathematics** twice in 2021 and 2022.
- Achieved rank among the **Top 1% of the state**, in the **National Standard Examination in Physics (NSEP) 2023**.
- Secured **State Rank 6 in Uttar Pradesh**, in the **National Talent Search Examination (NTSE) Stage 1**, 2021.
- Secured **7th position**, in **Stamatics IITK's Mathamania 2025**, featuring ~50 teams from various IITs.

PROGRAMMING ACHIEVEMENTS

- Peak Codeforces rating of **1781 - (Expert)**, handle [230478](#).
- Global 160th / India 40th** in Round 1033 and **Global 1029th / India 153rd** in Round 1026, among **21k official participants**.
- Secured **2nd place** for designing a decentralized task assignment system with a **Solidity smart contract**, featuring task dependencies, constraints, and gas-optimized execution for **Blockchain Problem Statement** at Takneek 2025.

WORK EXPERIENCE

Algoquant Fintech

Quantitative Analyst

May'26 - Jul'26

BSE Adapter: EOBI, ETI & NFOCAST streams

- Implemented a **UDP multicast** receiver and full **EOBI** binary parser, decoding multiple events like **ORDER_ADD/MODIFY/DELETE** from incremental and snapshot feeds.
- Built the **ETI** adapter end-to-end, implementing **80+ message types** on the wire from exchange to OMS, including the full **4-phase session logon**, **AES-256-GCM** encryption, and async callback handlers for order responses and execution reports.
- Designed modular ETI subsystems for **execution/fill classifier**, **ApplSeqNum/ApplMsgID gap-recovery** trackers, and a field-level **diagnostics framework** extending the MCX codebase.
- Implemented a **NFOCAST** receiver with a **FAST** decompression pipeline, decoding **Presence Maps** and dictionary-based operators to reconstruct top-of-book data with reduced bandwidth.

Optimizations & Benchmarking on ETI/MCX

- Eliminated per-message heap allocation in **AES-256-GCM** decryption by introducing a reusable buffer reinstating cleanup of completed order requests to prevent unbounded map growth reducing `shared_ptr` reference-counting overhead.
- Reduced steady-state ETI latency by gating wire-level diagnostics behind a **branch-predicted opt-in flag**, keeping the hot path free.
- Replaced dynamically allocated message buffers with **fixed-size arrays**, reducing allocation overhead.
- Optimized the order-fill path by collapsing a **double virtual-dispatch** lookup into a single **hash-map** access.

RELEVANT COURSES

Concurrent Programming
Generative AI
Probability & Statistics

Computer Architecture
GPU Architecture*
Discrete Mathematics

Parallel Computing*
Applied Cryptography
Linear Algebra and ODE

Programming for Performance*
Fundamentals of Computing
Architecture for Quantum Computing*

KEY PROJECTS

Optimising GPU Multiversion B-link Tree

Feb'26 - Apr'26

Course Project : CS636 (Analysis of Concurrency)

- Optimized a concurrent B-link tree data structure by **improving snapshot scalability** and **reducing kernel resource usage**.
- Profiled kernels** using **Nsight Compute** and **Nsight Systems**.
- Extended the snapshot mechanism to a **64-bit global counter** for improved correctness and **reduced register pressure** by **function refactoring** and **selective noinline annotations**.
- Reduced the hot versioned insert kernel from 69 to 57 registers/thread, increasing **theoretical occupancy upto 66.7%**.

Constant-Time Bitsliced AES-128

Mar'26 - Apr'26

Course Project : CS422 (Computer Architecture)

- Implemented a **constant-time, bitsliced AES-128** engine in C using the **Boyar-Peralta 128-gate circuit**, replacing S-box tables and processing 8 bit-planes concurrently via 32-bit operations.
- Precomputed the **key schedule** and optimized aligned 32-bit I/O to reduce load costs, achieving **59,763 cycles without AES-NI** and eliminating secret-dependent memory accesses and branches under leakage tests.
- Achieved **60% reduction in cycles over OpenSSL** (2.02M → 808K) **securing 1st rank in class**.

Instruction vs. CoT: Evaluating Performance

Oct'25 - Nov'25

Course Project : CS787 (Generative AI)

- Developed a **hybrid QA pipeline** combining a local **3B-parameter SLM** with **Gemini** for final multiple-choice reasoning, reducing reliance on expensive LLM inference.
- Designed a **decomposition pipeline** that generates targeted queries, retrieves snippets via the **Serper API**, and used the SLM to distill them into concise hints for Gemini.
- Improved CoT accuracy by **16 percentage points on HellaSwag** (71% → 87%) and **8 points on MUSR** (44% → 52%), while achieving **7 points on GPQA** over the Gemini-only baseline.

MAJOR COMPETITION

FedEx-Optimal Cargo Management for Flights

Nov'24 - Dec'24

InterIIT Tech Meet 13.0

- Designed an algorithm to place packages inside ULD containers considering **weight constraints** and **priority distribution**.
- Innovated **extreme points** algorithm in C++ to place packages inside ULD containers, thereby improving space utilisation.
- Developed **genetic algorithms**, used **simulated annealing** and **neighborhood swapping** methods to achieve **optimized cost**.
- Achieved the final volume packing efficiency of **83.02%**, thereby secured the **5th position for IIT Kanpur** out of all 23 IITs.

TECHNICAL SKILLS

- Languages & Tools:** C/C++, Python, Git/ Github, Intel PIN, Bash, GDB, CMake, MIPS Assembly, Verilog HDL, $\LaTeX$

POSITIONS OF RESPONSIBILITY

Secretary, Programming Club, IIT Kanpur

Jun'24 - Apr'25

- Organised the **Freshers' Programming Contest '24** and multiple introductory sessions on competitive programming.
- Organised **Spring Camp '25**, delivering lectures for **15 students** on topics such as competitive programming and blockchain.

(* ONGOING)